QUANT SERIES · PART 4 OF 4

# Stochastic Calculus from Zero

Part 4: rates and jumps, and the series capstone




Rates with Bip

STOCHASTIC CALCULUS FROM ZERO · PART 4 OF 4

# Stochastic Calculus from Zero

Part 4: rates and jumps, with Python and QuantLib

**Rates with Bip**

**Stochastic Calculus from Zero**. Part 4: rates and jumps, with Python and QuantLib.

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The AONIA curve in Chapters 1 and 3 is Series 1's illustrative curve (valued 25 September 2026), and the Hull-White parameters (mean reversion 3%, volatility 0.94%) are Series 4's illustrative calibration, rounded. The share and its jumps in Chapter 2 (100, risk-free rate 3.60%, diffusion volatility 15%, 0.5 jumps a year, mean log jump -10%, jump volatility 15%) are made up for teaching. Every path is simulated with numpy and a fixed seed. There is no market data.

The history in this book is checked against the original publications through their Crossref records (Hull and White 1990; Heath, Jarrow and Morton 1992; Merton 1976; Jamshidian 1989); sources are listed at the end of each chapter. Options are risky: a seller can lose far more than the premium received.

Every number in the text, tables and figures is computed by the companion Python notebooks (numpy, with QuantLib 1.43 as an independent check wherever a closed form exists), one per chapter, which reproduce each result. Every chapter's exercises are in its notebook too, and the code solutions check themselves.

This book is for education only. It is not financial, investment, legal or tax advice.

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# How to use this book

This book follows the three video episodes of Part 4 of *Stochastic Calculus from Zero* (Episodes 26 to 28), the last part of the series, one chapter per episode, with the proofs the videos only sketch. It assumes Parts 1 to 3: martingales and changes of measure, the Itô integral and Itô's formula, and the pricing tools of Part 3 (Girsanov, martingale representation, Feynman-Kac and change of numéraire).

Part 4 takes those tools to two places Part 3 didn't reach, and then uses all of them at once.

- **Chapter 1** fits today's yield curve. A short-rate model with constant parameters can't; Hull-White's time-dependent drift can, and the Heath-Jarrow-Morton framework explains why that drift is forced on us.
- **Chapter 2** adds jumps. Poisson processes, their compensators, Merton's jump-diffusion, the volatility skew it produces, and why jumps make the market incomplete.
- **Chapter 3** is the capstone: one option on an AONIA zero-coupon bond, priced four independent ways, with every route traced back to the chapter of the series it came from.

Each chapter ends with its one idea to remember (*Bip's rule*). The back matter has all fifteen exercises with full answers, the Part 4 checkpoint, a **glossary for the whole series** (the four parts' glossaries merged, each entry tagged with the part that introduces it) and further reading.

## Two running examples

**Today's curve** is Series 1's illustrative AONIA curve, valued 25 September 2026, from 3.85% overnight to 4.50% at 10 years, with Hull-White mean reversion 3% and volatility 0.94% (Series 4's illustrative calibration to AUD swaptions, rounded). **The share with jumps** is Series 3's share (100, risk-free rate 3.60%, a call with strike 105 expiring in a year) with a diffusion volatility of 15% and on average half a jump a year, each jump's log size normal with mean −10% and standard deviation 15%. Both are made up for teaching, and every simulation uses the seed 2026.

The notebooks are free at github.com/RatesWithBip/curriculum, in the `stochastic-calculus-from-zero` folder, each with an *Open in Colab* button. Each ends with its chapter's exercises, whose code solutions check themselves.

## Reading alongside

Steven Shreve's *Stochastic Calculus for Finance II*, Chapter 10 (term-structure models) and Chapter 11 (jumps), covers the same ground with full proofs; Damiano Brigo and Fabio Mercurio's *Interest Rate Models: Theory and Practice* is the reference for Hull-White and HJM; Rama Cont and Peter Tankov's *Financial Modelling with Jump Processes* for jumps. This book is written fresh from its own notebooks and uses none of their examples or exercises.

## Free, and staying free

The videos, the notebooks and this book are free. If they help you, you can support the channel with a coffee at ko-fi.com/rateswithbip.

This book is for education only and is not financial advice. Options are risky: a seller can lose far more than the premium received.

CHAPTER 1 · VIDEO EPISODE 26

# The Whole Curve

*A short-rate model with constant parameters can't reproduce today's curve. Hull-White's time-dependent drift can, and the HJM framework shows that drift is not a choice.*

## Before we start

The Part 3 checkpoint's answers are at the back of the Part 3 book and in the Episode 25 notebook. This chapter uses Part 3's Vasicek bond formula (Chapter 7 there) and the rule that discounted traded prices are martingales under the risk-neutral measure.

## The problem

Series 1's AONIA curve has zero rates rising from 3.92% at one year to 4.44% at ten. Part 3's Vasicek model, with its parameters ($r_0 = 3.85\%$, $\theta = 3.5\%$, $a = 0.30$, $\sigma = 1\%$), gives a falling curve that reaches 3.58%. Its bond prices $A(\tau)e^{-B(\tau)r_0}$ depend on four numbers, so it can produce only a four-parameter family of curves, and a curve built from a dozen market quotes is generally not in it. A model that misprices the plain bonds can't be trusted with options on them.

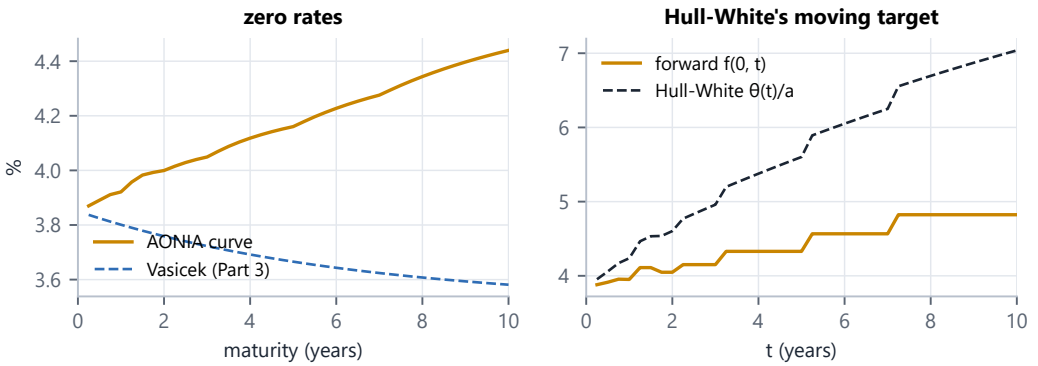



**Figure 1.1.** Left: zero rates from today's AONIA curve and from Part 3's Vasicek model. Right: today's instantaneous forward curve and Hull-White's moving target θ(t)/a. The steps come from the curve's piecewise-flat forwards.

## Hull-White

Hull and White (1990) keep Vasicek's form but let the target move:

$$dr = \big(\theta(t) - a\,r\big)\,dt + \sigma\,dW.$$

**Proposition.** With $f(0,t) = -\partial_t \ln P(0,t)$ today's instantaneous forward curve, the choice

$$\theta(t) = \frac{\partial f(0,t)}{\partial t} + a\,f(0,t) + \frac{\sigma^2}{2a}\big(1 - e^{-2at}\big)$$

makes the model's bond prices equal $P(0,T)$ for every $T$.

The proof comes out of the HJM framework below. With $a = 3\%$ and $\sigma = 0.94\%$:

Today's forward f(0, t) and the target θ(t)/a. With weak mean reversion the target sits well above the forward curve: the pull is gentle, so the target must aim high to lift rates along an upward-sloping curve.

| YEARS | FORWARD RATE, % | HULL-WHITE TARGET, % |
|---|---|---|
| 0.5 | 3.91 | 4.06 |
| 1.0 | 3.95 | 4.24 |
| 2.0 | 4.05 | 4.60 |
| 5.0 | 4.33 | 5.60 |
| 10.0 | 4.82 | 7.04 |

QuantLib's `HullWhite` model, given today's curve, prices zero-coupon bonds from 1 to 10 years equal to the curve's own discount factors; the largest gap is 1.1e-16.

## HJM: model every forward rate

Heath, Jarrow and Morton (1992) model the whole forward curve at once. For every maturity $T$,

$$df(t,T) = \alpha(t,T)\,dt + \sigma(t,T)\,dW_t,$$

starting from today's curve $f(0,T)$, with short rate $r(t) = f(t,t)$, bank account $B(t) = e^{\int_0^t r\,ds}$ and bond prices $P(t,T) = \exp\big(-\int_t^T f(t,u)\,du\big)$. Today's curve fits by construction. What is left to choose?

**Theorem (HJM drift condition).** $P(t,T)/B(t)$ is a martingale under $Q$ for every $T$ if and only if

$$\alpha(t,T) = \sigma(t,T)\int_t^T \sigma(t,u)\,du.$$

*Proof.* Let $Y_t = -\int_t^T f(t,u)\,du$, so $P = e^Y$. Differentiating under the integral (the lower limit contributes $f(t,t) = r(t)$),

$$dY = r\,dt - \Big(\int_t^T \alpha(t,u)\,du\Big)dt - \Big(\int_t^T \sigma(t,u)\,du\Big)dW.$$

Write $\Sigma(t,T) = \int_t^T \sigma(t,u)\,du$. By Itô, $dP/P = dY + \tfrac{1}{2}(dY)^2$, so

$$\frac{dP}{P} = \Big(r - \int_t^T \alpha(t,u)\,du + \tfrac{1}{2}\Sigma^2\Big)dt - \Sigma\,dW.$$

$P/B$ is a martingale exactly when the drift of $dP/P$ is $r$ (Part 3, Chapter 5), that is $\int_t^T \alpha(t,u)\,du = \tfrac{1}{2}\Sigma(t,T)^2$ for every $T$. Differentiating in $T$ gives $\alpha(t,T) = \sigma(t,T)\,\Sigma(t,T)$; integrating back gives the converse. □

You choose the volatility; arbitrage chooses the drift. For constant $\sigma(t,T) = \sigma$, $\alpha = \sigma^2(T-t)$: the **Ho-Lee** model.

## Hull-White is HJM

**Proposition.** With $\sigma(t,T) = \sigma e^{-a(T-t)}$, the HJM short rate is $r(t) = \varphi(t) + x(t)$, where $dx = -ax\,dt + \sigma\,dW$, $x(0) = 0$, and

$$\varphi(t) = f(0,t) + \frac{\sigma^2}{2a^2}\big(1 - e^{-at}\big)^2;$$

this is Hull-White with the $\theta(t)$ above.

*Proof.* Integrating the forward-rate SDE at $T = t$,

$$r(t) = f(t,t) = f(0,t) + \int_0^t \alpha(s,t)\,ds + \int_0^t \sigma e^{-a(t-s)}\,dW_s.$$

The last integral is $x(t)$, the Ornstein-Uhlenbeck process started at 0 (Part 3, Chapter 2). With the HJM drift $\alpha(s,t) = \frac{\sigma^2}{a}e^{-a(t-s)}\big(1 - e^{-a(t-s)}\big)$, the middle integral is $\frac{\sigma^2}{2a^2}(1 - e^{-at})^2$. So $r = \varphi + x$ and

$$dr = \varphi'(t)\,dt - ax\,dt + \sigma\,dW = \big(\varphi'(t) + a\varphi(t) - ar\big)\,dt + \sigma\,dW.$$

Expanding $\varphi' + a\varphi$, the two convexity terms combine as $(1 - e^{-at})^2 + 2e^{-at}(1 - e^{-at}) = 1 - e^{-2at}$, which gives the $\theta(t)$ of the proposition. Because HJM starts from today's curve, Hull-White with this $\theta$ reprices it. □

## The drift is not a choice

Simulate the forward curve weekly for 2 years, on a weekly grid of maturities to 10 years, with Hull-White's volatility, and check the martingale property $E[D(0,2)\,P(2,10)] = P(0,10)$, where $D(0,2) = e^{-\int_0^2 r\,dt}$. The same 20,000 sets of random draws are used twice: with the HJM drift and with no drift.

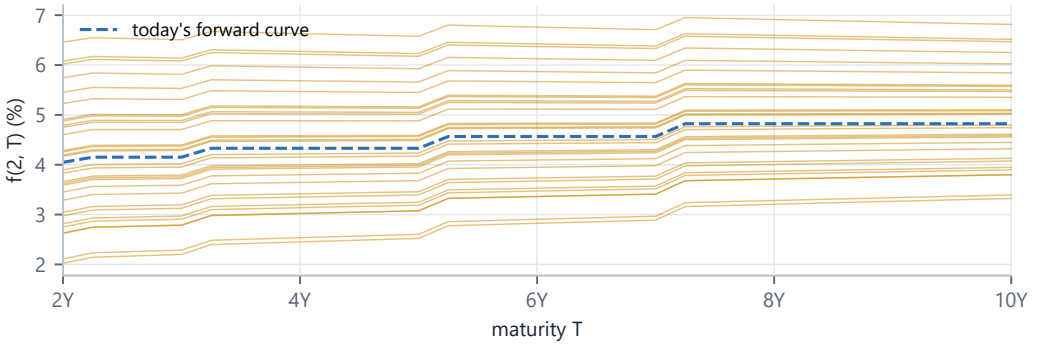



**Figure 1.2.** Thirty simulated forward curves f(2, T), two years from now, around today's forward curve.

| DRIFT | $E[D(0,2)\,P(2,10)]$ | VERSUS TODAY, BP OF PRICE |
|---|---|---|
| Today's $P(0,10)$ on the grid | 0.64153 | |
| HJM drift | 0.64140 ± 0.00048 | -2.0 |
| No drift | 0.64494 | 53.2 |

With the HJM drift the discounted bond is a martingale, within its standard error. Without it, the same draws overprice the bond by 53 basis points; on the same draws the difference is 0.00354 with a standard error of only 2.6e-06. That is an arbitrage.

## A common mistake

"In HJM you choose the drift." You choose the volatility of each forward rate; no-arbitrage chooses the drift.

**Bip's rule.** Choose the volatility; arbitrage chooses the drift.

**Try this.** If every forward rate has the same constant volatility $\sigma$, what is the HJM drift?

## Sources

Companion notebook: `s5e26_hull_white_hjm.ipynb`. Every number in this chapter is computed by it.


1. The Econometric Society / JSTOR (Crossref record), *D. Heath, R. Jarrow and A. Morton, Bond pricing and the term structure of interest rates: a new methodology for contingent claims valuation, Econometrica 60 (1992), 77-105*. https://doi.org/10.2307/2951677
2. Oxford University Press (Crossref record), *J. Hull and A. White, Pricing interest-rate-derivative securities, Review of Financial Studies 3 (1990), 573-592*. https://doi.org/10.1093/rfs/3.4.573
3. QuantLib, *QuantLib 1.43: HullWhite short-rate model fitted to a term structure (discountBond)*. https://www.quantlib.org/

CHAPTER 2 · VIDEO EPISODE 27

# Jumps

*A Poisson process counts jumps, and subtracting its expected count makes it a martingale. Merton's jump-diffusion prices a call in closed form, bends the smile into a skew, and breaks delta hedging.*

## Last chapter's question

The answer: $\alpha(t,T) = \sigma \int_t^T \sigma\, du = \sigma^2(T-t)$, the Ho-Lee model.

## The Poisson process

A **Poisson process** with intensity $\lambda$ is a counting process $N_t$ with $N_0 = 0$, jumps of size 1, and independent increments with $N_t - N_s \sim \text{Poisson}\big(\lambda(t-s)\big)$:

$$P(N_t - N_s = k) = e^{-\lambda(t-s)}\,\frac{\big(\lambda(t-s)\big)^k}{k!}.$$

So $E[N_t] = \text{Var}(N_t) = \lambda t$ (Exercise 2.1) and $P(N_t = 0) = e^{-\lambda t}$.

**Proposition.** The **compensated** process $\widetilde{N}_t = N_t - \lambda t$ is a martingale.

*Proof.* For $s < t$, $N_t - N_s$ is independent of $\mathcal{F}_s$ with mean $\lambda(t-s)$, so $E[N_t - \lambda t \mid \mathcal{F}_s] = N_s + \lambda(t-s) - \lambda t = N_s - \lambda s$. □

$\lambda t$ is the **compensator** of $N$: the part you can predict. Unlike Brownian motion, $\widetilde{N}$ has quadratic variation $N_t$ (each jump contributes $1^2$), not $t$, and its paths jump. In the simulation with $\lambda = 0.5$ over five years, $N_5$ averages 2.494 with variance 2.490, against $\lambda t = 2.5$.

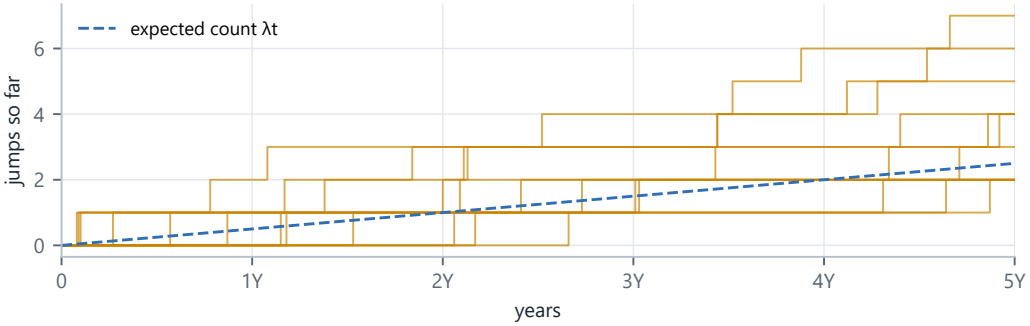



**Figure 2.1.** Twelve simulated Poisson paths, half a jump a year, and the expected count λt.

## Merton's jump-diffusion

Merton (1976) adds jumps to geometric Brownian motion. Under the risk-neutral measure,

$$\frac{dS_t}{S_{t-}} = (r - \lambda\bar{k})\,dt + \sigma\,dW_t + (J-1)\,dN_t,$$

where at each jump the price is multiplied by an independent $J$ with $\ln J \sim N(\mu_J, \delta^2)$, and $\bar{k} = E[J-1] = e^{\mu_J + \frac{1}{2}\delta^2} - 1 =$ -8.5%. The solution is

$$S_T = S_0 \exp\left(\left(r - \lambda\bar{k} - \tfrac{1}{2}\sigma^2\right)T + \sigma W_T + \sum_{i=1}^{N_T} \ln J_i\right).$$

**Why $-\lambda\bar{k}$.** Over $dt$ the jump term adds $(J-1)\,dN$, whose expectation is $\bar{k}\lambda\,dt$. The compensated jump term $(J-1)\,dN - \lambda\bar{k}\,dt$ has mean zero, so with drift $r - \lambda\bar{k}$ the total expected return is $r$ and $e^{-rt}S_t$ is a martingale. The simulated forward, 103.67, matches $100e^{0.036}$.

## The price, by conditioning on the jumps

**Proposition (Merton's formula).** With $\lambda' = \lambda(1+\bar{k})$, $\sigma_n^2 = \sigma^2 + n\delta^2/T$ and $r_n = r - \lambda\bar{k} + n\ln(1+\bar{k})/T$,

$$C = \sum_{n=0}^{\infty} e^{-\lambda' T}\frac{(\lambda' T)^n}{n!}\,\mathrm{BS}(S_0, K, T, r_n, \sigma_n).$$

*Proof.* Given $N_T = n$, $\ln S_T$ is normal with variance $\sigma^2 T + n\delta^2 = \sigma_n^2 T$ and $E[S_T \mid N_T = n] = S_0 e^{(r-\lambda\bar{k})T}(1+\bar{k})^n = S_0 e^{r_n T}$. For a lognormal $S_T$ with forward $S_0 e^{r_n T}$ and total variance $\sigma_n^2 T$,

$$e^{-rT}E\left[(S_T - K)^+ \mid N_T = n\right] = e^{(r_n - r)T}\,\mathrm{BS}(S_0, K, T, r_n, \sigma_n),$$

because Black-Scholes at rate $r_n$ discounts at $r_n$ instead of $r$. Averaging over $P(N_T = n) = e^{-\lambda T}(\lambda T)^n/n!$ and using $e^{(r_n - r)T} = e^{-\lambda \bar{k} T}(1+\bar{k})^n$, the weights combine into $e^{-\lambda' T}(\lambda' T)^n/n!$. □

| METHOD | CALL, STRIKE 105 |
|---|---|
| Merton's formula | 6.835 |
| QuantLib (Bates, variance frozen) | 6.835 |
| Monte Carlo, 400,000 paths | 6.834 ± 0.018 |
| Black-Scholes at the same total volatility | 7.264 |

The Python build of QuantLib 1.43 has no Merton engine; Bates's model (Heston's variance plus Merton's jumps) with the variance frozen at $\sigma^2$ is Merton's model, and agrees to $10^{-5}$. The total volatility $\sqrt{\sigma^2 + \lambda(\mu_J^2 + \delta^2)} = 19.7\%$ matches the variance but not the shape: Black-Scholes at that volatility overprices this slightly out-of-the-money call.

## The skew

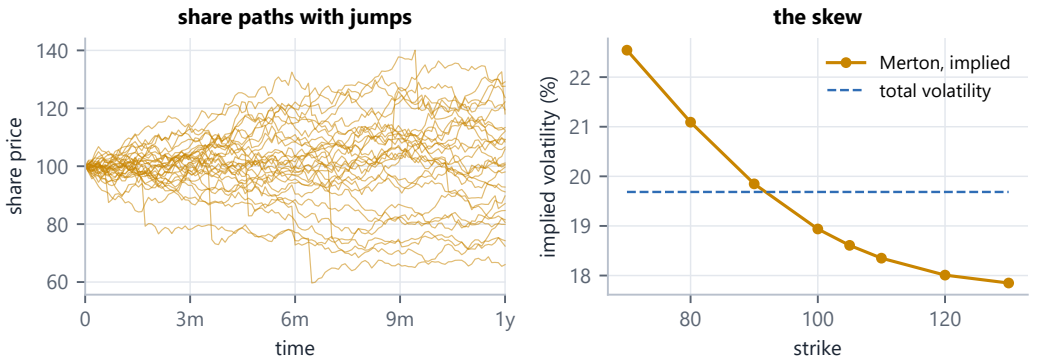



**Figure 2.2.** Left: thirty share paths with jumps. Right: the Black-Scholes implied volatility of Merton prices across strikes, against the total volatility.

The implied volatility falls from 22.5% at strike 70 to 17.8% at 130. Downward jumps fatten the left tail of $S_T$, so out-of-the-money puts (low strikes) are worth more than any single lognormal allows: a **skew**, as in Series 3's equity smile. Upward jumps tilt it the other way (Exercise 2.4).

## Hedging breaks

Sell the call and delta-hedge with the share. Between two rebalancing dates, a Brownian move of size $\sqrt{\Delta t}$ costs about $\frac{1}{2}\Gamma S^2\sigma^2\big(\Delta t - (\Delta W)^2\big)$, which averages out and shrinks as $\Delta t \to 0$ (Part 3, Chapter 6). A jump moves the share by a finite amount however small $\Delta t$ is, and the option's curvature turns it into a loss of order $\frac{1}{2}\Gamma(\Delta S)^2$ that no delta can anticipate.

Standard deviation of the hedging error at expiry, 2,000 paths each. No jumps: geometric Brownian motion with the same total volatility.

| REBALANCING | ERROR SPREAD: NO JUMPS | ERROR SPREAD: WITH JUMPS |
|---|---|---|
| monthly | 1.98 | 3.98 |
| weekly | 0.99 | 3.66 |
| daily | 0.46 | 3.61 |

## Incomplete markets

With one Brownian motion and one traded share, every payoff can be replicated (Part 3, Chapter 6). Here there are two sources of risk. Martingales now need two integrands, one against $dW$ and one against the compensated jump measure, and the share can supply only one. The market is **incomplete**: many equivalent martingale measures price the share correctly, they differ in how they price jump risk (the intensity and jump distribution under $Q$ can differ from those under $P$), and the option's price depends on that choice. Merton's formula assumes jump risk is not priced. In practice, traders hedge jump risk with other options.

## A common mistake

"Hedge often enough and the risk goes away." True for Brownian risk; false across a jump. Daily hedging here still leaves an error spread of 3.61.

**Bip's rule.** You can't delta-hedge a jump.

**Try this.** With half a jump a year on average, what is the chance of no jump in a year?

## Sources

Companion notebook: `s5e27_jumps_merton.ipynb`. Every number in this chapter is computed by it.

1. Elsevier (Crossref record), *R. C. Merton, Option pricing when underlying stock returns are discontinuous, Journal of Financial Economics 3 (1976), 125-144*. https://doi.org/10.1016/0304-405X(76)90022-2
2. QuantLib, *QuantLib 1.43: BatesProcess with BatesEngine (Heston variance plus Merton jumps)*. https://www.quantlib.org/

CHAPTER 3 · VIDEO EPISODE 28

# One AUD Bond Option, Four Ways

*A one-year call on a five-year AONIA zero-coupon bond under Hull-White, priced by a closed form, two Monte Carlos and a PDE. Four routes, one number, and every tool from the series.*

## Last chapter's question

The answer is 60.7%: $P(N_1 = 0) = e^{-0.5}$.

## The trade

On Series 1's AONIA curve, $P(0,1) = 0.96155$ and $P(0,5) = 0.81219$. The option is a European call expiring at $T_1 = 1$ on the zero-coupon bond maturing at $T_2 = 5$, struck at the forward bond price

$$K = F_0 = \frac{P(0,5)}{P(0,1)} = 0.84467.$$

The model is Chapter 1's Hull-White, fitted to the curve: $r = x + \varphi(t)$, $dx = -ax\,dt + \sigma\,dW$, $a = 3\%$, $\sigma = 0.94\%$. At expiry the bond is a known function of $x(1)$,

$$P(1,5) = \frac{P(0,5)}{P(0,1)} \exp\Big(\tfrac{1}{2}\big(V(1,5) - V(0,5) + V(0,1)\big) - B(1,5)\,x(1)\Big),$$

with $B(t,T) = (1 - e^{-a(T-t)})/a$ and $V(t,T)$ the variance of $\int_t^T x\,du$ given $x(t)$. So each route only has to get the distribution of $x(1)$, and the discounting, right.

## Route 1: the closed form

**Proposition.** Under the 1-year forward measure, $F_t = P(t,5)/P(t,1)$ is a lognormal martingale with total variance

$$v^2 = \frac{\sigma^2}{a^2}\big(1 - e^{-a(T_2 - T_1)}\big)^2\,\frac{1 - e^{-2aT_1}}{2a},$$

so the call is $C = P(0,T_1)\big(F_0 N(d_1) - K N(d_2)\big)$ with $d_{1,2} = \big(\ln(F_0/K) \pm \tfrac{1}{2}v^2\big)/v$.

*Proof.* $F$ is a traded price in units of the numéraire $P(t,1)$, so it is a martingale under $Q^1$ (Part 3, Chapter 9). Each bond has volatility $-\sigma B(t,T)$, so $d\ln F$ has volatility $-\sigma\big(B(t,5)-B(t,1)\big) = -\frac{\sigma}{a}e^{-a(1-t)}\big(1-e^{-4a}\big)$, which is deterministic: $\ln F_1$ is normal. Integrating the squared volatility over $[0,1]$ gives $v^2$ (Exercise 3.1). A lognormal martingale with total variance $v^2$ has the Black call price; since $P(1,1)=1$, $V_0 = P(0,1)\,E^1[(F_1-K)^+]$. $\square$

Here $v = 3.49\%$ and $C = 113.10$ basis points of face value, the same as QuantLib's `HullWhite.discountBondOption` to $10^{-10}$. This is Jamshidian's (1989) formula.

## Route 2: risk-neutral paths

Under $Q$ (Part 3, Chapter 5), $C = E_Q\big[e^{-\int_0^1 r\,dt}\,(P(1,5)-K)^+\big]$. The pair $\big(x(1), \int_0^1 x\,dt\big)$ is jointly normal, since both are Itô integrals of deterministic functions (Part 2): $x(1) = \sigma\int_0^1 e^{-a(1-s)}dW_s$ and $\int_0^1 x\,dt = \sigma\int_0^1 B(s,1)\,dW_s$. By the Itô isometry, their variances and covariance are

$$\operatorname{Var} x(1) = \frac{\sigma^2}{2a}(1-e^{-2a}), \qquad \operatorname{Cov} = \frac{\sigma^2}{2a^2}(1-e^{-a})^2, \qquad \operatorname{Var}\int_0^1 x = V(0,1),$$

so one exact step simulates both. The discount factor is $e^{-\int_0^1 r} = P(0,1)\,e^{-\frac12 V(0,1) - \int_0^1 x\,dt}$; across the paths it averages $0.96154$, against $P(0,1) = 0.96155$. With 200,000 paths, $C = 113.36 \pm 0.38$ bp.

## Route 3: forward-measure paths

Under $Q^1$ (Part 3, Chapter 9), $C = P(0,1)\,E^1\big[(P(1,5)-K)^+\big]$: discount once. By Girsanov with the bond's volatility, $dx = \big(-ax - \sigma^2 B(t,1)\big)\,dt + \sigma\,dW^1$, so $x(1)$ is still normal with the same variance, and its mean shifts to

$$-\frac{\sigma^2}{a^2}\big(1-e^{-a}\big) + \frac{\sigma^2}{2a^2}\big(1-e^{-2a}\big) = -0.43 \text{ bp}.$$

$C = 113.11 \pm 0.38$ bp. On the same paths the bond averages $0.84471$, the forward price: the martingale property again.

## Route 4: the PDE

By Feynman-Kac (Part 3, Chapter 7), the option's value $u(t,x)$ before expiry solves

$$u_t - a\,x\,u_x + \tfrac12\sigma^2 u_{xx} - \big(x+\varphi(t)\big)\,u = 0, \qquad u(1,x) = \big(P(1,5;x)-K\big)^+,$$

with $\varphi(t) = f(0,t) + \frac{\sigma^2}{2a^2}(1 - e^{-at})^2$ (Chapter 1). Crank-Nicolson on $x \in [-15\%, 15\%]$ with 401 points and 400 time steps, stepped back from expiry, gives $u(0,0) = 113.08$ bp.

## Four routes, one number

| ROUTE | PRICE (BP OF FACE) | TOOLS | FROM |
|---|---|---|---|
| Closed form = QuantLib | 113.10 | Numéraire, lognormal martingale, Black | Part 3, Chapters 5, 9 |
| Risk-neutral paths | 113.36 ± 0.38 | Itô integrals, isometry, OU, discounting inside | Part 2; Part 3, Chapters 2, 5 |
| Forward-measure paths | 113.11 ± 0.38 | Girsanov, change of numéraire | Part 3, Chapters 5, 9 |
| PDE | 113.08 | Feynman-Kac, finite differences | Part 3, Chapter 7 |

The largest gap to the closed form is 0.26 bp, inside the simulation noise. Four calculations that share nothing but the model agree, which tests the theory and the code at once.

## The whole series in one page

| PART | IDEA | WHERE IT WAS USED TODAY |
|---|---|---|
| 1 | Conditional expectation, martingales, changing the odds | Every price is a conditional expectation under a chosen measure |
| 2 | Brownian motion, the Itô integral, Itô's formula | The joint normal law of $x$ and $\int x$; bond volatilities |
| 3 | SDEs, Girsanov, Feynman-Kac, numéraires | Routes 1 to 4 |
| 4 | Fitting today's curve; jumps | The Hull-White fit; and what this model leaves out |

What this model leaves out matters too: one factor moves the whole curve together, and there are no jumps on RBA meeting dates. Two-factor models, and jump-diffusions for rates, are where to go next.

## A common mistake

"Stochastic calculus is only for exotic options." This chapter priced a plain option on a zero-coupon bond on Australia's overnight-rate curve. Caps, swaptions, XVA and every delta hedge rest on the same tools.

**Bip's rule.** One number, many roads: if they agree, you understand it.

**Try this.** The Part 4 checkpoint is at the back of the book.

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### Sources

Companion notebook: `s5e28_capstone.ipynb`. Every number in this chapter is computed by it.


1. Wiley (Crossref record), *F. Jamshidian, An exact bond option formula, Journal of Finance 44 (1989), 205-209*. https://doi.org/10.1111/j.1540-6261.1989.tb02413.x
2. QuantLib, *QuantLib 1.43: HullWhite short-rate model fitted to a term structure (discountBondOption)*. https://www.quantlib.org/

# Exercises with answers

Five exercises per chapter: three by hand and two in code. They are the same exercises as at the end of each chapter's notebook, where the code solutions run and check themselves; try each before reading its answer. "Section" in an exercise means a section of that chapter's notebook.

## Chapter 1: The Whole Curve

**Exercise 1.1** *(by hand).* Derive the HJM drift condition. Write $Y_t = -\int_t^T f(t,u)\,du$, so $P = e^Y$; find $dY$, then require the drift of $P/B$ to vanish.

*Answer.* $dY = f(t,t)\,dt - \big(\int_t^T \alpha\,du\big)dt - \big(\int_t^T \sigma\,du\big)dW$. By Itô, $dP/P = dY + \frac{1}{2}(dY)^2$, so the drift of $P/P$ is $r - \int_t^T \alpha\,du + \frac{1}{2}\big(\int_t^T \sigma\,du\big)^2$. For $P/B$ to be a martingale this must equal $r$: $\int_t^T \alpha\,du = \frac{1}{2}\big(\int_t^T \sigma\,du\big)^2$. Differentiate in $T$: $\alpha(t,T) = \sigma(t,T)\int_t^T \sigma(t,u)\,du$.

**Exercise 1.2** *(by hand).* Show that $\sigma(t,T) = \sigma e^{-a(T-t)}$ gives the HJM drift $\frac{\sigma^2}{a}e^{-a(T-t)}\big(1 - e^{-a(T-t)}\big)$.

*Answer.* $\int_t^T \sigma e^{-a(u-t)}\,du = \frac{\sigma}{a}\big(1 - e^{-a(T-t)}\big)$; multiply by $\sigma e^{-a(T-t)}$.

**Exercise 1.3** *(by hand).* Why can't Part 3's Vasicek model (constant $\theta$) fit an arbitrary curve?

*Answer.* Its bond prices $A(\tau)e^{-B(\tau)r_0}$ depend on four numbers $(r_0, a, \theta, \sigma)$, so it can only produce a four-parameter family of curves. A curve built from twelve market quotes generally isn't in that family; a time-dependent $\theta(t)$ gives one free number per date.

**Exercise 1.4** *(in code).* For a flat 4% curve, compute $\theta(t)/a$ at 1, 5 and 10 years. Where does the extra over 4% come from?

Solution 1.4, from the notebook

```
lv = [0.04 + SIG_HW**2 / (2 * A_HW**2) * (1 - math.exp(-2 * A_HW * t)) for t in (1, 5, 10)]
assert all(l > 0.04 for l in lv)
print([f"{l:.4%}" for l in lv], "- the convexity term sigma^2/(2a^2)(1 - e^(-2at)) lifts the
level above the forward")
```

*What it prints.* `['4.2859%', '5.2723%', '6.2148%'] - the convexity term sigma^2/(2a^2) (1 - e^(-2at)) lifts the level above the forward`

**Exercise 1.5** *(in code)*. Rerun the HJM check with half the HJM drift. Where does the answer land?

Solution 1.5, from the notebook

```
f = np.tile(f_today, (n_paths, 1)); logD = np.zeros(n_paths)
for k in range(steps):
    tau = grid[k:] - k * h
    sig = SIG_HW * np.exp(-A_HW * tau)
    logD -= f[:, k] * h
    f[:, k:] += 0.5 * sig * SIG_HW * (1 - np.exp(-A_HW * tau)) / A_HW * h + np.outer(dW[k],
sig)
half = float((np.exp(logD) * np.exp(-(f[:, steps:-1] * h).sum(axis=1))).mean())
assert hjm["hjm"] < half < hjm["zero"]
print(f"full drift {hjm['hjm']:.5f}, half {half:.5f}, none {hjm['zero']:.5f}; today's P(0,10)
{target:.5f}")
```

*What it prints.* `full drift 0.64140, half 0.64317, none 0.64494; today's P(0,10) 0.64153`

## Chapter 2: Jumps

**Exercise 2.1** *(by hand)*. Show that $E[N_t] = \mathrm{Var}(N_t) = \lambda t$ from the Poisson probabilities $P(N_t = k) = e^{-\lambda t}(\lambda t)^k / k!$.

*Answer.* $E[N] = \sum k\, e^{-m} m^k / k! = m \sum e^{-m} m^{k-1}/(k-1)! = m$ with $m = \lambda t$. Similarly $E[N(N-1)] = m^2$, so $\mathrm{Var}(N) = m^2 + m - m^2 = m$.

**Exercise 2.2** *(by hand)*. Why does Merton's drift include $-\lambda \bar{k}$?

*Answer.* Over $dt$ a jump arrives with probability $\lambda\, dt$ and multiplies $S$ by $J$, adding $E[J-1] = \bar{k}$ per jump: an expected return of $\lambda \bar{k}\, dt$. For $e^{-rt}S$ to be a martingale the total expected return must be $r\, dt$, so the drift is $r - \lambda \bar{k}$.

**Exercise 2.3** *(by hand)*. What does Merton's formula reduce to when $\lambda = 0$? And when $\delta = 0$ and $\mu_J = 0$?

*Answer.* $\lambda = 0$: only the $n = 0$ term survives, which is Black-Scholes with volatility $\sigma$. $\delta = \mu_J = 0$: $J = 1$, the jumps don't move the price, $\bar{k} = 0$, and every term is the same Black-Scholes price; the weights sum to 1.

**Exercise 2.4** *(in code)*. Recompute the smile with a positive mean jump, $\mu_J = +10\%$. Which way does it tilt?

Solution 2.4, from the notebook

```
iv = {k: bs_implied_vol(float(merton_call(S0, k, T, R, VOL, LAM, 0.10, DJ)), S0, k, T, R)
for k in (80, 100, 120)}
assert iv[120] > iv[80]
print({k: f"{v:.2%}" for k, v in iv.items()}, "- upward jumps fatten the right tail: high
strikes now need more volatility")
```

*What it prints.* `{80: '18.21%', 100: '19.30%', 120: '21.10%'} - upward jumps fatten the right tail: high strikes now need more volatility`

**Exercise 2.5** *(in code)*. Hedge daily with smaller jumps (mean log jump −5%, jump volatility 5%). How does the error floor change?

Solution 2.5, from the notebook

```
small = float(hedge(252, True, muj=-0.05, dj=0.05).std())
assert small < out["hedge"]["daily_jump"]
print(f"daily hedging error: {out['hedge']['daily_jump']:.2f} with the base jumps,
{small:.2f} with small jumps")
```

*What it prints.* `daily hedging error: 3.61 with the base jumps, 0.86 with small jumps`

## Chapter 3: One AUD Bond Option, Four Ways

**Exercise 3.1** *(by hand)*. Derive the volatility of $F_t = P(t,5)/P(t,1)$ in Hull-White: show $d\ln F$ has volatility $\sigma\big(B(t,5) - B(t,1)\big)$, and integrate its square from 0 to 1.

*Answer.* Each bond has volatility $-\sigma B(t,T)$, so the ratio has $-\sigma(B(t,5) - B(t,1)) = -\frac{\sigma}{a}e^{-a(1-t)}(1 - e^{-4a})$. Squaring and integrating over $[0,1]$: $\frac{\sigma^2}{a^2}(1 - e^{-4a})^2\frac{1-e^{-2a}}{2a}$, the square of the $v$ in Section 4.

**Exercise 3.2** *(by hand)*. Show put-call parity for the bond option: call − put $= P(0,5) - K\,P(0,1)$. With the strike at the forward, what does that make the put?

*Answer.* Call − put pays $P(1,5) - K$ at time 1, worth $P(0,5) - KP(0,1)$. With $K = P(0,5)/P(0,1)$ this is 0: the put equals the call.

**Exercise 3.3** *(by hand)*. For a *Bermudan* option (exercisable on several dates), which of the four routes still works directly, and why?

*Answer.* The PDE (or a tree): stepping backwards, at each exercise date replace $u$ by $\max(u, \text{exercise value})$. The closed form has no early exercise, and plain forward Monte Carlo doesn't know the continuation value (it needs regression, as in Series 4's AMC).

**Exercise 3.4** *(in code)*. Price the matching put four ways: QuantLib, Black on the forward, forward-measure paths and parity.

Solution 3.4, from the notebook

```
ql_put = hw.discountBondOption(ql.Option.Put, Kb, T1, T2)
black_put = P01 * (Kb * float(norm_cdf(-(d1 - v))) - F0 * float(norm_cdf(-d1)))
mc_put = P01 * np.maximum(Kb - bond_f, 0)
parity = ql_call - (P05 - Kb * P01)
se = mc_put.std() / math.sqrt(n)
assert abs(ql_put - black_put) < 1e-10 and abs(ql_put - parity) < 1e-10 and
abs(mc_put.mean() - ql_put) < 3 * se
print(f"put: QuantLib {ql_put * 1e4:.2f} bp, Black {black_put * 1e4:.2f}, parity {parity *
1e4:.2f}, "
      f"forward-measure paths {mc_put.mean() * 1e4:.2f} ± {se * 1e4:.2f}")
```

*What it prints.* `put: QuantLib 113.10 bp, Black 113.10, parity 113.10, forward-measure paths 112.75 ± 0.36`

**Exercise 3.5** *(in code).* A caplet on the 3-month simple forward rate from 1 year to 1.25 years, strike 4%, pays $\tau \max(L - K, 0)$ at 1.25 years. Show it equals $(1 + K\tau)$ puts on $P(1, 1.25)$ with strike $1/(1 + K\tau)$, by risk-neutral paths.

Solution 3.5, from the notebook

```
tau, Kc = 0.25, 0.04
P_1_125 = hw_bond(T1, T1 + tau, x1, P0, A_HW, SIG_HW)
L = (1 / P_1_125 - 1) / tau                                  # the simple forward rate fixed at
1 year
caplet_mc = disc * P_1_125 * tau * np.maximum(L - Kc, 0)   # paid at 1.25: discount to 1 by
P(1, 1.25)
caplet_bond = (1 + Kc * tau) * hw.discountBondOption(ql.Option.Put, 1 / (1 + Kc * tau), T1,
T1 + tau)
se = caplet_mc.std() / math.sqrt(n)
assert abs(caplet_mc.mean() - caplet_bond) < 3 * se
print(f"caplet: paths {caplet_mc.mean() * 1e4:.3f} ± {se * 1e4:.3f} bp; (1 + K tau) x bond
put {caplet_bond * 1e4:.3f} bp")
```

*What it prints.* `caplet: paths 10.377 ± 0.031 bp; (1 + K tau) x bond put 10.387 bp`

# Checkpoint, glossary and further reading

## Part 4 checkpoint

Try these first; the answers follow.

1. For a constant forward-rate volatility $\sigma(t,T) = 0.01$, write the HJM drift.
2. A Poisson process has $\lambda = 2$ a year. What are the mean and variance of $N_3$, and $P(N_1 = 0)$?
3. Why can't a delta hedge remove jump risk? What extra instrument would help?
4. Name the four routes to the capstone price, and the chapter of the series each relies on.

**Answers.**

1. $\alpha(t,T) = 0.01 \times 0.01\,(T-t) = 0.0001\,(T-t)$: Ho-Lee (Chapter 1).
2. Mean and variance $2 \times 3 = 6$; $P(N_1 = 0) = e^{-2} = 13.5\%$ (Chapter 2).
3. A jump moves the share by a finite amount between rebalancing dates, and the option's curvature turns it into a loss of order $\frac{1}{2}\Gamma(\Delta S)^2$ that no delta anticipates. A second option, whose value also responds to jumps, can offset it (Chapter 2).
4. The closed form (Black on the forward bond price: Part 3, Chapter 9), risk-neutral paths (Part 3, Chapter 5), forward-measure paths (Part 3, Chapter 9) and the PDE (Feynman-Kac: Part 3, Chapter 7) (Chapter 3).

## Glossary of the whole series

Every term from the four books, in one list; the tag says which part introduces it (where two parts define a term, the later wording is used).

**Adapted.** A process whose value at time *t* is known at time *t*. *(Parts 1, 2)*

**Affine model.** A short-rate model whose bond prices are $A(\tau)e^{-B(\tau)r}$; Vasicek and CIR are affine. *(Part 3)*

**Barrier option.** An option that starts (knock-in) or stops (knock-out) existing if the underlying reaches a barrier. *(Part 3)*

**Brownian bridge.** Brownian motion conditioned on its value at the end of an interval. *(Part 3)*

**Brownian motion (Wiener process).** Starts at 0; independent, normal increments with variance equal to the elapsed time; continuous paths. *(Part 2)*

**Central limit theorem.** A sum of many independent pieces, standardised, is approximately normal. *(Part 1)*

**Cholesky factorisation.** Writing a correlation matrix as $LL^\top$; multiplying independent Brownian motions by $L$ gives correlated ones. *(Part 2)*

**CIR model.** $dr = a(\theta - r)\,dt + \sigma\sqrt{r}\,dW$ (Cox, Ingersoll and Ross). *(Part 3)*

**Compensator.** The predictable part of a counting process: $\lambda t$ for a Poisson process, so $N_t - \lambda t$ is a martingale. *(Part 4)*

**Complete market.** One in which every payoff can be replicated by trading, so it has one price. *(Part 3)*

**Conditional expectation.** $E[X \mid \mathcal{F}_n]$: the average of $X$ within each block of information; the least-squares best guess of $X$ using what is known at time $n$. *(Part 1)*

**Cross variation.** The limit of $\sum \Delta X\,\Delta Y$; for correlated Brownian motions, $\rho t$. *(Part 2)*

**Donsker's theorem.** Scaled random walks converge to Brownian motion as whole paths: the functional central limit theorem. *(Part 2)*

**Equivalent measures.** Two probabilities that agree on which events are impossible. *(Part 1)*

**Euler (Euler-Maruyama) scheme.** $X_{n+1} = X_n + \mu\,\Delta t + \sigma\,\Delta W_n$. *(Part 3)*

**Event.** A set of worlds. Its probability is their total weight. *(Part 1)*

**Expectation.** The weighted average of a random variable over all worlds. *(Part 1)*

**Exponential martingale.** $\exp(\sigma W_t - \frac{1}{2}\sigma^2 t)$. *(Part 2)*

**Feller's condition.** $2a\theta \geq \sigma^2$: a CIR process never reaches zero. *(Part 3)*

**Feynman-Kac theorem.** A discounted expected payoff solves $u_t + \mu u_x + \frac{1}{2}\sigma^2 u_{xx} - ru = 0$. *(Part 3)*

**Filtration.** A sequence of sigma-algebras $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \ldots$: information that grows and is never forgotten. *(Part 1)*

**Forward measure.** The measure with a zero-coupon bond $P(t,T)$ as numéraire; prices in units of that bond are martingales. *(Part 3)*

**Forward rate curve.** $f(t,T)$, the instantaneous rate for borrowing at $T$ agreed at $t$; $P(t,T) = e^{-\int_t^T f(t,u)\,du}$. *(Part 4)*

**Geometric Brownian motion.** $dS = \mu S\,dt + \sigma S\,dW$; $S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$. *(Part 3)*

**Girsanov's theorem.** Under $dQ/dP = e^{-\theta W_T - \frac{1}{2}\theta^2 T}$, $W_t + \theta t$ is a Brownian motion. *(Part 3)*

**Half-life.** The time for the expected gap to the long-run level to halve: $\ln 2 / a$. *(Part 3)*

**HJM drift condition.** $\alpha(t,T) = \sigma(t,T)\int_t^T \sigma(t,u)\,du$: in the Heath-Jarrow-Morton framework no-arbitrage fixes the drift of every forward rate. *(Part 4)*

**Ho-Lee model.** HJM with constant forward-rate volatility: drift $\sigma^2(T-t)$. *(Part 4)*

**Hull-White model.** $dr = (\theta(t) - ar)\,dt + \sigma\,dW$ with $\theta(t)$ chosen to reprice today's curve; HJM with $\sigma(t,T) = \sigma e^{-a(T-t)}$. *(Part 4)*

**Incomplete market.** One with more sources of risk than traded assets to hedge them; prices are no longer unique. *(Part 4)*

**Independent increments.** The change of a process over a period is independent of its past. True of the random walk and of Brownian motion. *(Part 1)*

**Itô integral.** The mean-square limit of left-point sums $\sum H_{t_i}\,\Delta W_i$ for an adapted $H$. *(Part 2)*

**Itô isometry.** $E[(\int H\,dW)^2] = E[\int H^2\,dt]$. *(Part 2)*

**Itô process.** $dX = \mu\,dt + \sigma\,dW$ with adapted $\mu$ and $\sigma$. *(Part 2)*

**Itô-Doeblin formula.** $df(t,X) = f_t\,dt + f_x\,dX + \frac{1}{2}f_{xx}\,(dX)^2$, with $(dX)^2 = \sigma^2\,dt$. *(Part 2)*

**Jamshidian's formula.** The bond option price in Vasicek and Hull-White: Black's formula on the forward bond price. *(Part 4)*

**Jensen's inequality.** For convex $f$, $E[f(X)] \geq f(E[X])$. *(Part 2)*

**Jump-diffusion (Merton).** Geometric Brownian motion plus Poisson jumps that multiply the price by a lognormal factor; drift $r - \lambda\bar{k}$ under $Q$. *(Part 4)*

**Known at time n (measurable).** An event that is a union of time-$n$ blocks; a random variable constant on each block. *(Part 1)*

**Law of large numbers.** The average of many independent copies of a random variable settles on its expectation. *(Part 1)*

**Lipschitz condition.** $|f(x) - f(y)| \le K|x - y|$; with linear growth, it guarantees a unique strong solution. *(Part 3)*

**Lévy's characterisation.** A continuous martingale from 0 with quadratic variation $t$ is a Brownian motion. *(Part 2)*

**Market price of risk.** $(\mu - r)/\sigma$: extra drift per unit of volatility. *(Part 3)*

**Markov property.** The future depends on the past only through the present. *(Part 1)*

**Martingale.** A process with $E[M_{n+1} \mid \mathcal{F}_n] = M_n$: the best guess for tomorrow is today. A **submartingale** drifts up ($\ge$), a **supermartingale** down ($\le$). *(Part 1)*

**Martingale representation theorem.** Every square-integrable Brownian martingale is $M_0 + \int H\, dW$. *(Part 3)*

**Milstein scheme.** Euler plus $\frac{1}{2}\sigma\sigma'((\Delta W)^2 - \Delta t)$. *(Part 3)*

**Multiplication table.** $dW \cdot dW = dt$, $dW \cdot dt = dt \cdot dt = 0$, $dW_1 \cdot dW_2 = \rho\, dt$. *(Part 2)*

**Novikov's condition.** $E[e^{\frac{1}{2}\int_0^T \theta^2 dt}] < \infty$: enough for Girsanov with a random $\theta$. *(Part 3)*

**Numéraire.** A positive traded asset used as the unit of value. *(Part 3)*

**Ornstein-Uhlenbeck process.** $dX = a(\theta - X)\, dt + \sigma\, dW$. *(Part 3)*

**Poisson process.** A counting process with independent Poisson increments: $E[N_t] = \mathrm{Var}(N_t) = \lambda t$. *(Part 4)*

**Quadratic variation.** The limit of $\sum(\Delta X)^2$; $[W]_t = t$, $[\sigma W]_t = \sigma^2 t$. *(Parts 1, 2)*

**Radon-Nikodym derivative.** $Z = dQ/dP$, the ratio of two equivalent probabilities, world by world. $E_P[Z] = 1$ and $E_Q[X] = E_P[ZX]$. *(Part 1)*

**Random variable.** A number attached to each world: a function on the sample space. *(Part 1)*

**Random walk.** The sum of independent $\pm 1$ steps. *(Part 1)*

**Realised volatility.** The square root of the sum of squared log returns over a period, annualised. *(Part 2)*

**Reflection principle.** $P(\max_{t \le T} W_t \ge m) = 2P(W_T \ge m)$. *(Part 3)*

**Risk-neutral probability.** The odds $Q$ under which discounted prices are martingales; a pricing tool, not a forecast. *(Part 1)*

**Running maximum.** $M_T = \max_{t \le T} W_t$. *(Part 3)*

**Sample space.** The set $\Omega$ of all possible worlds. *(Part 1)*

**SDE (stochastic differential equation).** $dX = \mu(t, X)\,dt + \sigma(t, X)\,dW$, meaning the corresponding Itô integral equation. *(Part 3)*

**Self-similarity.** A Brownian path, zoomed in by $c$ in time and $\sqrt{c}$ in height, is again a Brownian path. *(Part 2)*

**Sigma-algebra.** A collection of events closed under complements and countable unions; with finitely many worlds, the unions of the blocks of a partition. *(Part 1)*

**Simple integrand.** A process constant on each step of a partition, known at the start of the step. *(Part 2)*

**Stratonovich integral.** The limit of middle-point sums; keeps the school chain rule but is not a martingale. *(Part 2)*

**Strong and weak error.** $E|X_N - X_T|$ (paths) and $|E[g(X_N)] - E[g(X_T)]|$ (averages). *(Part 3)*

**Total variation (length).** The limit of $\sum |\Delta X|$; infinite for Brownian motion. *(Part 2)*

**Tower property.** $E\big[E[X \mid \mathcal{F}_n] \mid \mathcal{F}_m\big] = E[X \mid \mathcal{F}_m]$ for $m \le n$. *(Part 1)*

**Variance.** $E[(X - E[X])^2]$; for independent random variables, variances add. *(Part 1)*

**Vasicek model.** The Ornstein-Uhlenbeck process as a short-rate model. *(Part 3)*

**Volatility skew.** Implied volatility falling with strike; downward jumps produce one. *(Part 4)*

## Further reading

This book is written from its own notebooks; these are excellent companions. None of their examples or exercises are used here.

- Steven E. Shreve, *Stochastic Calculus for Finance II: Continuous-Time Models*: Chapter 10 (term-structure models, HJM) for Chapters 1 and 3, Chapter 11 (jumps) for Chapter 2.
- Damiano Brigo and Fabio Mercurio, *Interest Rate Models: Theory and Practice*: Chapter 3 for Hull-White, Chapter 5 for HJM.

- Rama Cont and Peter Tankov, *Financial Modelling with Jump Processes*: Poisson processes, Merton's model and incompleteness.
- Paul Glasserman, *Monte Carlo Methods in Financial Engineering*: Chapter 3 for simulating Hull-White and HJM.

Check the chapter pointers against the edition you have.

## Where next

This is the end of *Stochastic Calculus from Zero*. *Options from Zero* (Series 3) uses these tools on the volatility smile, local and stochastic volatility; *XVA: The Price After the Price* (Series 4) uses Hull-White exposures for counterparty risk. Both are free at the same repository.